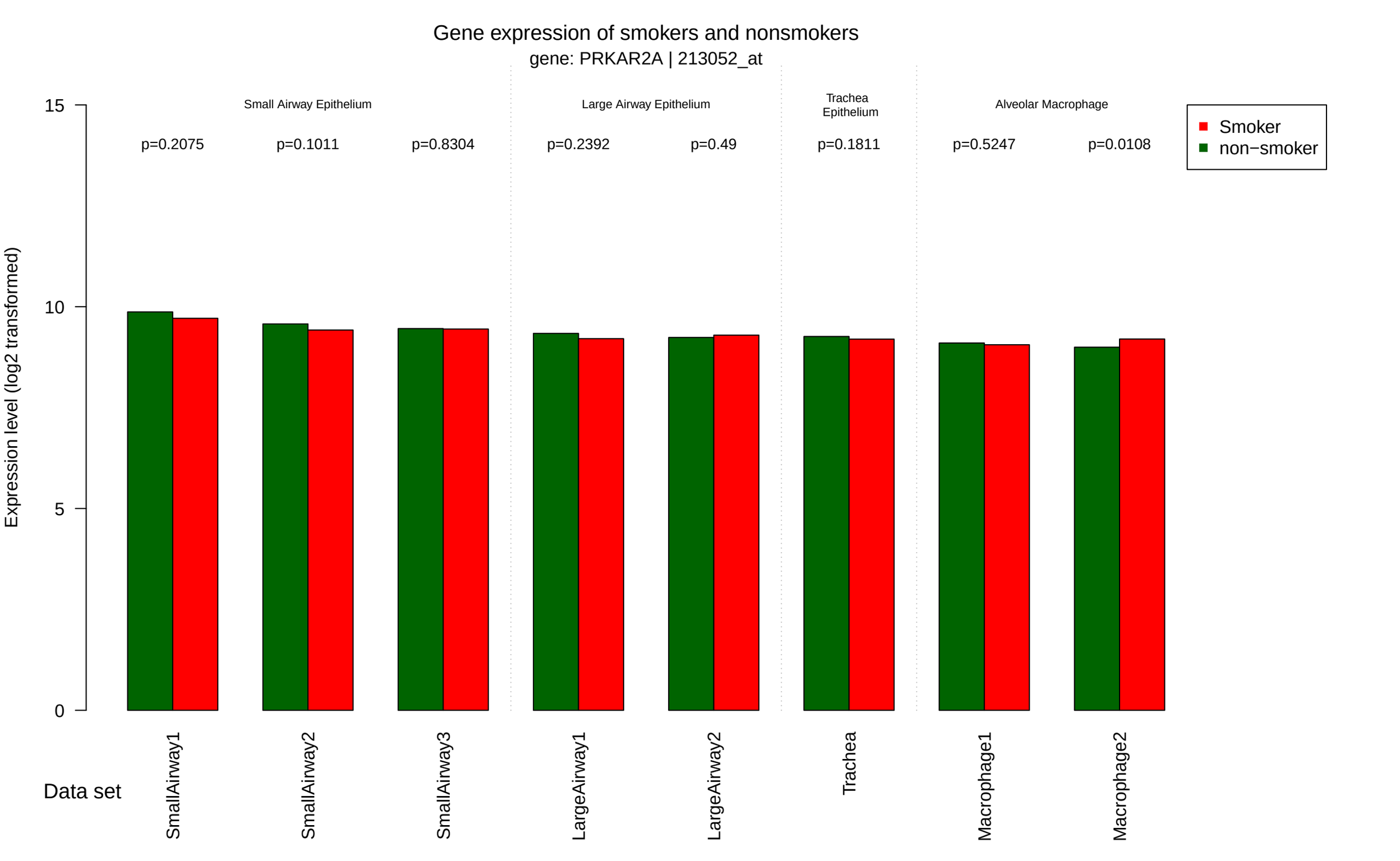
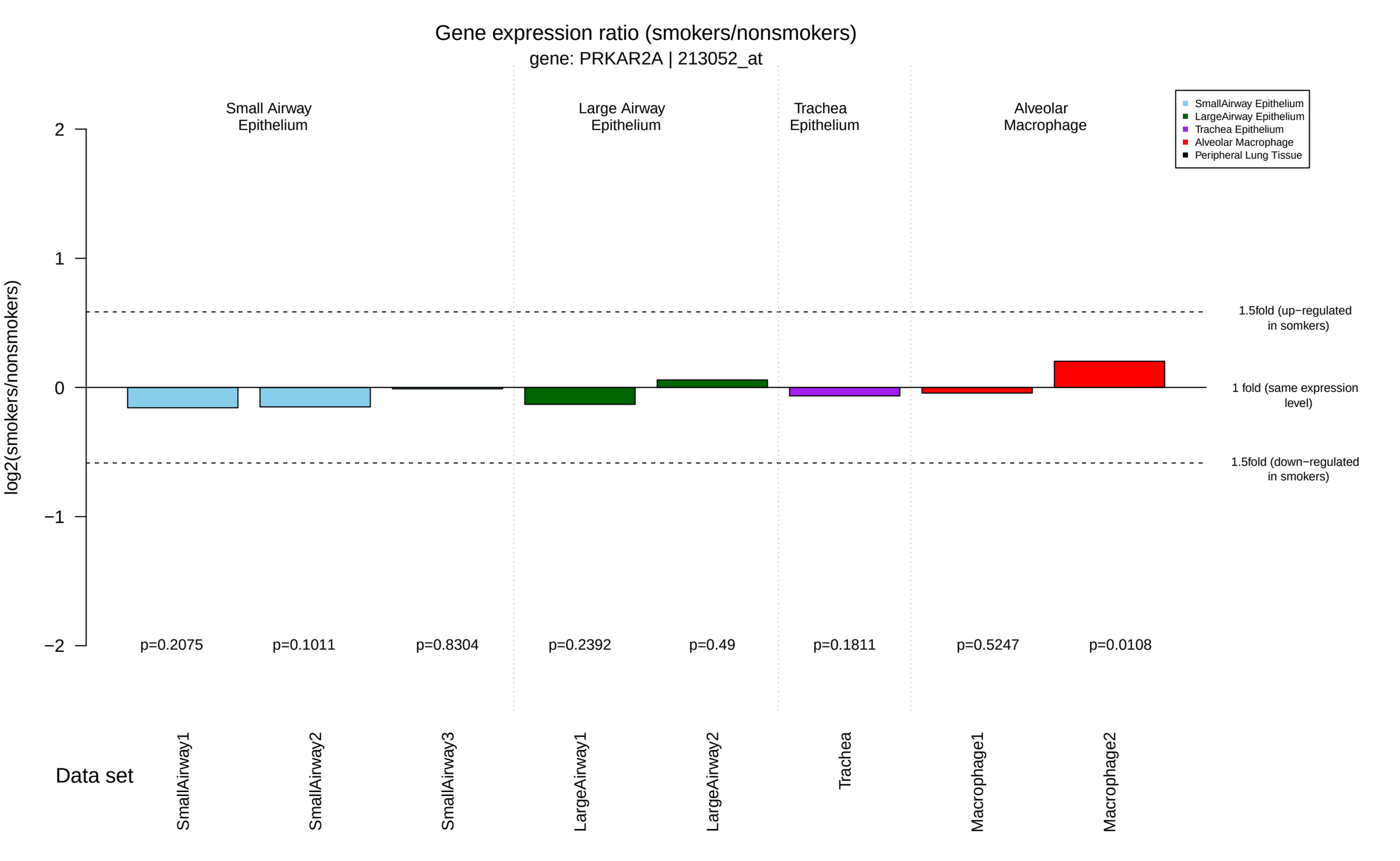
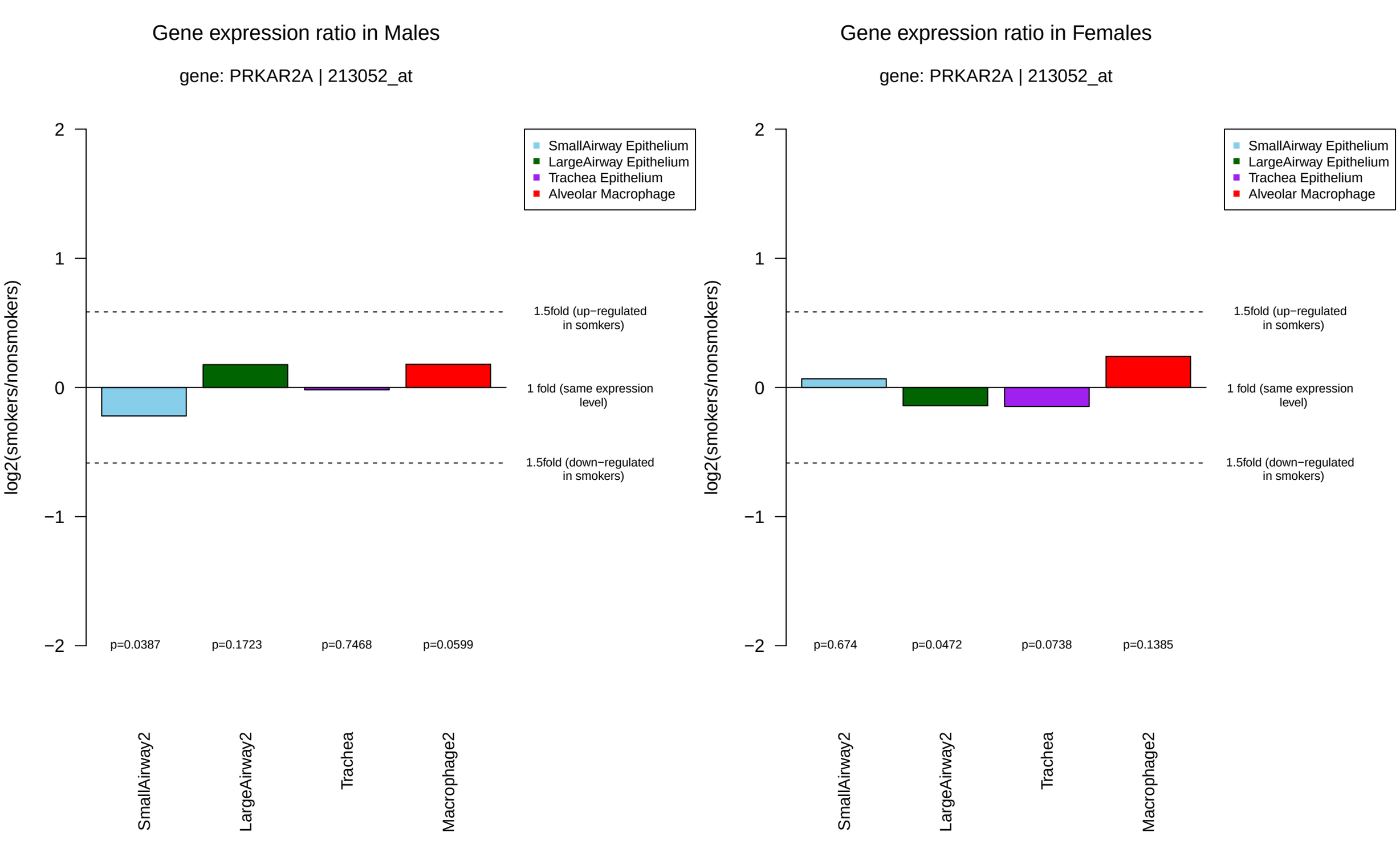
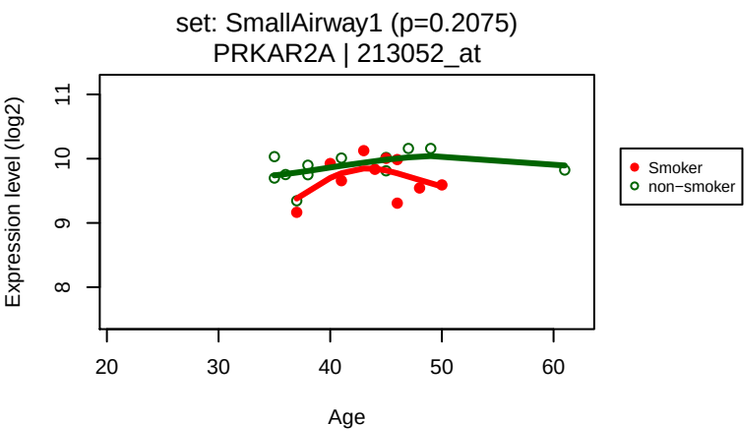
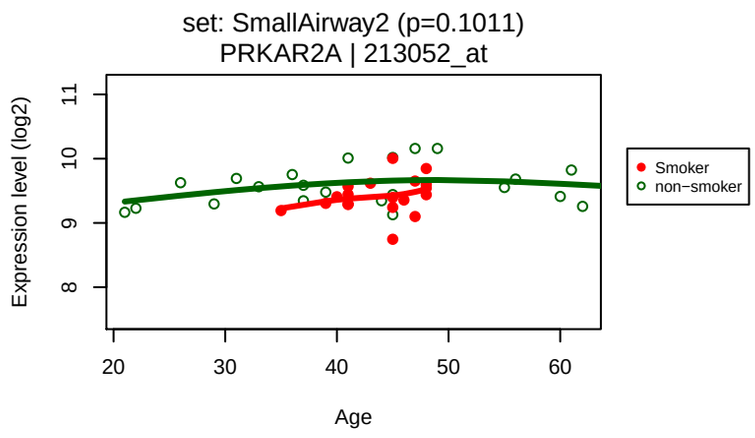
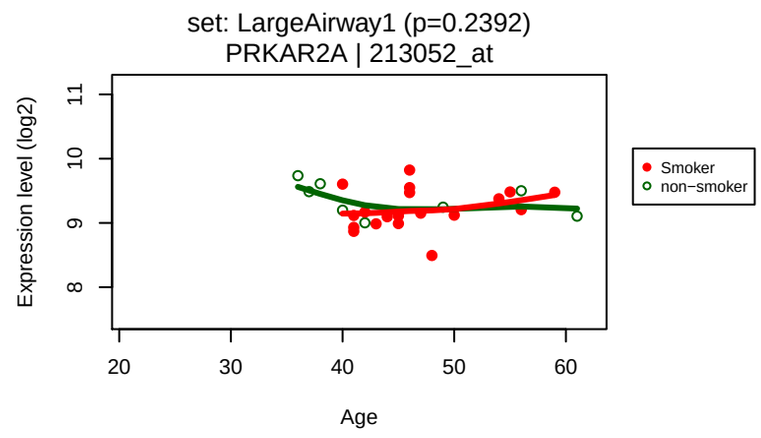

Gene expression of smokers and nonsmokers
gene: PRKAR2A | 213052_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.2075
p=0.1011
p=0.8304
p=0.2392
p=0.49
p=0.1811
p=0.5247
p=0.0108
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

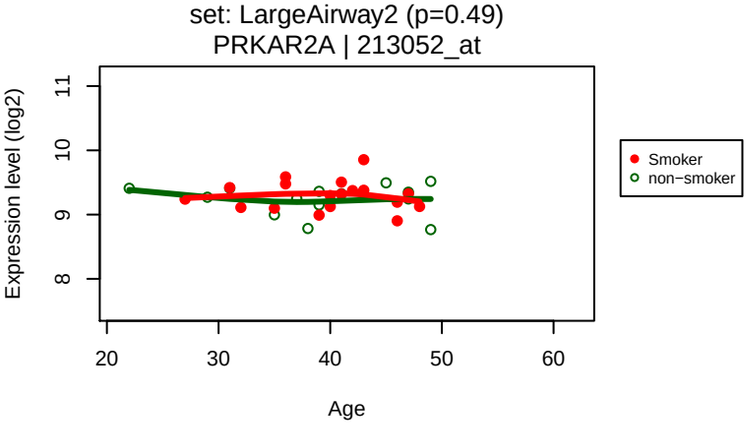

Gene expression ratio (smokers/nonsmokers)
gene: PRKAR2A | 213052_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2075
p=0.1011
p=0.8304
p=0.2392
p=0.49
p=0.1811
p=0.5247
p=0.0108
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

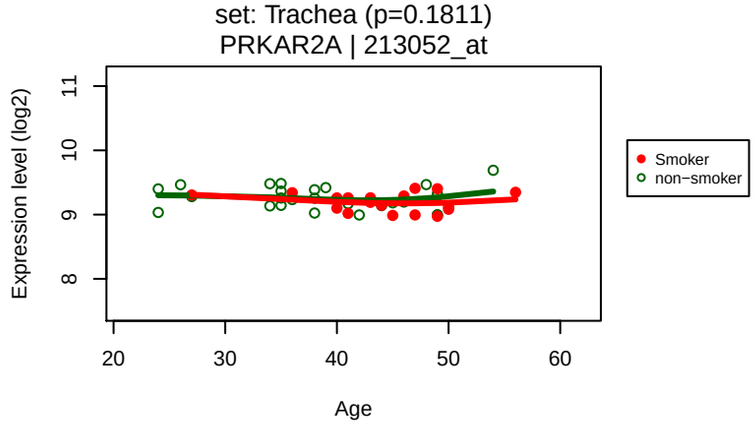

Gene expression ratio in Males
gene: PRKAR2A | 213052_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0387
p=0.1723
p=0.7468
p=0.0599
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: PRKAR2A | 213052_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.674
p=0.0472
p=0.0738
p=0.1385
SmallAirway2
LargeAirway2
Trachea
Macrophage2

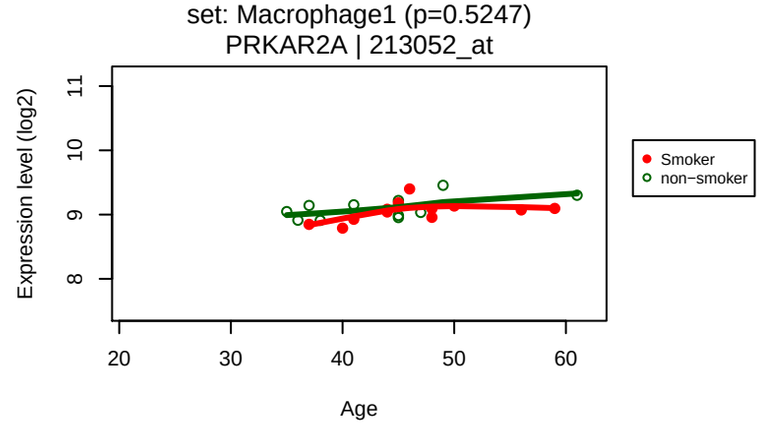

set: SmallAirway1 (p=0.2075)
PRKAR2A | 213052_at
Expression level (log2)
Age
Smoker
non−smoker

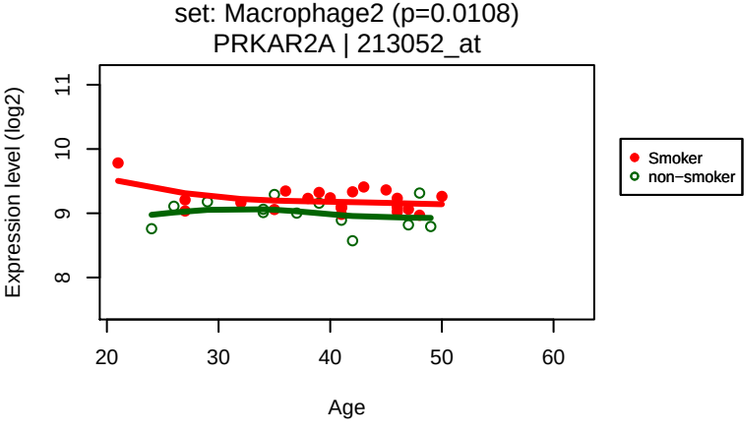

set: SmallAirway2 (p=0.1011)
PRKAR2A | 213052_at
Expression level (log2)
Age
Smoker
non−smoker

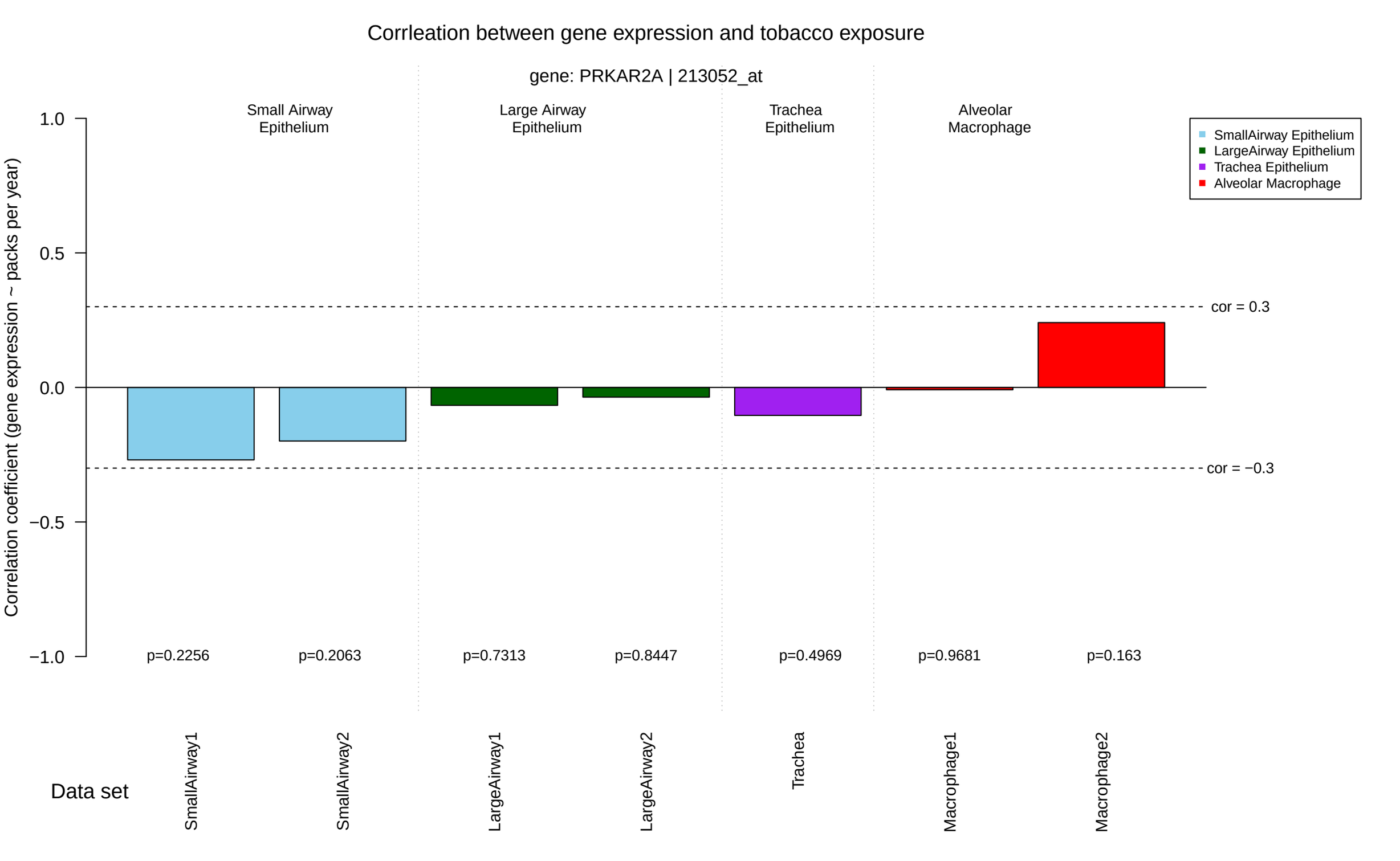

set: LargeAirway1 (p=0.2392)
PRKAR2A | 213052_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.49)
PRKAR2A | 213052_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.1811)
PRKAR2A | 213052_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.5247)
PRKAR2A | 213052_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.0108)
PRKAR2A | 213052_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: PRKAR2A | 213052_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
Correlation coefficient (gene expression ~ packs per year)
p=0.2256
p=0.2063
p=0.7313
p=0.8447
p=0.4969
p=0.9681
p=0.163
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2